A Large Language Model Paradigm Integrating Expert Diagnostic Data for Fetal Congenital Heart Disease Screening and Prenatal Diagnosis in Low-Resourced Healthcare Settings

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Abstract

To reduce inequities in prenatal care, digital medicine must move beyond proof of concept toward context-adapted, auditable and scalable implementation. To address the reliance of fetal congenital heart disease (FCHD) screening on specialist expertise, we developed ExpEvaFCHD, a large language model (LLM) workflow that integrates local expert diagnostic evidence. Using structured measurements acquired with ultrasound equipment, the system combines field alignment, similar-case retrieval and evidence aggregation to generate traceable diagnostic interpretations without fine-tuning the base model, supporting physician interpretation and patient-facing explanations. Using 50,093 expert-consensus cases, we evaluated six base-model backends in a prospective single-center cohort (n = 1,458) and a multicenter cohort (n = 3,370) from 15 hospitals across five Chinese provinces, spanning four levels of care: internationally leading specialist centers, provincial centers, regional hospitals and primary-level facilities. Compared with direct generation by the base models, ExpEvaFCHD increased sensitivity from 53.24%–73.24% to 88.82%–92.94% and specificity from 24.69%–79.96% to 98.03%–98.93% in prospective validation. In multicenter validation, sensitivity and specificity reached 85.76%–89.87% and 97.18%–98.10%, respectively. AI-assisted interpretation increased physician sensitivity to 93.82% and 91.77% in the two cohorts, respectively, and reduced report drafting and review time by approximately 4.8 min on average. With identical evidence, the full Qwen3-Max system and template reports yielded identical binary classification results, while the full system produced more readable and complete reports. ExpEvaFCHD provides a clinically actionable, context-adapted and scalable approach to implementing digital medicine for prenatal screening in low-resourced settings using existing clinical data.

Introduction

Improving the efficiency and equity of public health systems is a global challenge amid population growth, aging and limited resources. In many countries, medical deserts and other inequalities in access to specialist care continue to delay diagnosis and widen disparities, particularly where travel burdens, workforce shortages and uneven access to technology coexist.[1,2] Fetal congenital heart disease (FCHD) exemplifies this implementation challenge. Prenatal diagnosis of critical congenital heart disease is associated with lower preoperative mortality

and better coordinated perinatal care, yet detection remains uneven across institutions and populations.3 The central question is therefore not only whether digital tools can improve diagnostic performance in principle, but also whether they can extend specialist-level interpretive support to prenatal care settings with limited fetal cardiology expertise.

Recent research in fetal cardiology and obstetric ultrasound has demonstrated substantial progress in artificial intelligence (AI) for view recognition, measurement, detection of suspicious findings, screening for congenital abnormalities, portable-device support and care pathways enabled by tele-ultrasound.[4–18] However, the dominant technical approach remains image-centered: many systems depend on curated video sequences or large annotated image datasets, require substantial retraining or site-specific adaptation, and are difficult to implement where image archives are fragmented, data governance is restrictive or computing budgets are limited.[5,8–10,13] Even when image-based systems perform well, translation into community practice remains affected by domain shift, equipment heterogeneity, operator variability and the practical constraints of deployment in relatively low-resourced settings.[8,9,15–18]

Large language models (LLMs) offer another route to improving access to care by conveying structured clinical knowledge, supporting patient communication and guiding workflows through natural-language interaction at relatively low marginal cost. Emerging health-system applications have used LLMs to classify patient-reported symptoms and service needs at scale. Recent perspectives have also called for real-world trials of LLM-supported clinical decision tools in African primary care rather than reliance on laboratory demonstrations.[19,20] Meanwhile, systematic ethical reviews identify equity, privacy, transparency and harmful misinformation as persistent concerns when LLMs are introduced into medicine and healthcare.[21]

Improved capabilities alone are insufficient to support clinical deployment. General-purpose LLM systems such as ChatGPT and Gemini demonstrate broad medical knowledge and flexibility across tasks, but their outputs remain susceptible to hallucinations, unstable calibration and discrepancies between fluent wording and clinically valid reasoning.[22,23] Critiques of these foundational issues further emphasize that electronic health record and medical-language applications remain highly dependent on data quality, nonclinical artifacts of documentation and task formulation, making external grounding and human evaluation essential.[24,25] Even in a randomized clinical trial, providing physicians with LLM outputs did not reliably improve their diagnostic reasoning for complex cases.[26]

In response, the field has increasingly moved from stand-alone chatbots toward domain-grounded architectures. Outside fetal medicine, an integrated image–language system for primary diabetes care suggests that clinically useful improvements are most credible when generation is coupled with task-specific signals, structured evidence and clearly scoped evaluation.[27] Quantitative analyses of visual–textual medical diagnosis likewise indicate that semantic alignment alone does not ensure robust clinical inference, further underscoring the need for explicit evidence pathways.[28] Related work in medical question answering and terminology standardization also suggests that retrieval-linked external knowledge and constrained reasoning can improve controllability, citation quality and interpretability.[29–31]

These advances do not directly resolve the deployment bottlenecks that are critical to structured prenatal diagnostic support. Publicly available medical corpora largely comprise educational materials, isolated case reports or general question-answering resources rather than large, disease-specific, expert-validated structured datasets. Heterogeneity in measurement names, units and missingness across institutions introduces naming ambiguity and data-quality problems that can destabilize retrieval and inference. Privacy requirements and resource constraints in county-level, community and mobile healthcare settings also reduce the feasibility of workflows that depend on cloud services, repeated fine-tuning or image-intensive inference.[21,24]

The central challenge is therefore not simply to train more capable models, but to provide clinically meaningful decision support under local institutional control, with clearly defined governance and realistic infrastructure requirements. Routine fetal echocardiography offers a practical opportunity because many clinically decisive features are already available as device-measured structured parameters: cardiac chamber dimensions, outflow vessel diameters, Doppler velocities, cardiac axis and cardiothoracic ratio. Compared with free-text reports or raw images, these measurements are easier to standardize across centers and can anchor retrieval and reasoning in an auditable feature space. We therefore developed ExpEvaFCHD (Expert Evaluation of Fetal Congenital Heart Disease Risk), an evidence-constrained workflow grounded in local structured evidence that combines deterministic column-wise standardization, similarity retrieval from an institution-controlled case database, polarity-controlled evidence aggregation and evidence-aligned constrained generation. The system serves two interconnected user groups: frontline ultrasound physicians requiring structured decision support for FCHD screening and preliminary interpretation, and pregnant women and their families requiring transparent, accessible explanations grounded in the same structured evidence rather than opaque model outputs. ExpEvaFCHD differs from generic retrieval-augmented generation (RAG) in three ways relevant to implementation in low-resourced settings. Its inputs are structured fetal echocardiographic measurements obtained directly from ultrasound equipment, rather than raw images or unrestricted narrative text. Retrieval operates in a standardized measurement space, providing greater robustness to cross-center documentation heterogeneity than text-only retrieval. The expert case database, retrieval index and rule-based evidence checks remain under local institutional control, while the generation backend can use open-weight models or API services according to available resources. We evaluated ExpEvaFCHD using an expert database containing 50,093 cases, a prospective single-center cohort and a multicenter validation cohort spanning 15 hospitals. The evaluation focused on whether the framework could support clinical action, accommodate local infrastructure constraints and maintain sufficient stability for scaling across centers.

Results

Study cohorts and case composition

We constructed and evaluated ExpEvaFCHD using three complementary cohorts (Table 1). FetalCardioDB, the local expert retrieval database, comprised 50,093 independent fetal echocardiography cases collected at Beijing Anzhen Hospital between January 2016 and November 2025. All diagnostic labels were established by consensus among FCHD specialists. Prospective single-center and multicenter validation were both conducted from December 1, 2025 to March 1, 2026. The prospective cohort included 1,458 cases, comprising 340 positive and 1,118 negative cases. The multicenter cohort included 3,370 cases from 15 hospitals in Anhui, Fujian, Guangdong, Liaoning and Ningxia, spanning internationally leading specialist centers, provincial centers, regional hospitals and primary-level facilities; 316 cases were positive and 3,054 were negative. The proportions of positive cases were 23.32% and 9.38%, respectively. The retrieval database and validation cohorts were separated chronologically. Unique patient identifiers were used to exclude duplicate records, and repeat visits and referrals involving multicenter patients were checked to ensure that no patients overlapped across the three cohorts. Provincial case composition is presented in Table 2, and the geographical distribution of participating hospitals is shown in Supplementary Fig. S6.

Table 1 | Case composition and maternal–fetal characteristics of the three study cohorts

Diagnostic family	Database n	Maternal age years	Gestational age weeks	Prospective n	Maternal age years	Gestational age weeks	Multicenter n	Maternal age years	Gestational age weeks
Negative/normal	42,461	31.5 (17–54)	26.4 (17.4–41.1)	1118	32.37（19–46）	25.69（15.9–39.4）	3054	28.78（19–48）	23.53（15.0–36.3）
Septal defects	3,374	31.0 (16–45)	26.7 (17.4–37.6)	139	31.12（20–44）	26.04（16.4–38.3）	139	30.03（19–49）	25.50（16.3–38.0）
Great vessels/aortic arch	2,957	30.8 (16–46)	26.3 (16.0–37.4)	116	31.21（18–45）	26.01（16.4–38.3）	123	29.51（19–40）	26.10（17.0–38.0）
Venous return/systemic veins	1,866	30.8 (16–46)	27.2 (16.6–36.4)	47	31.96（24–41）	26.37（16.4–36.3）	34	28.86（23–38）	25.52（19.0–30.4）
Valves/outflow tracts	1,652	31.1 (18–48)	26.0 (16.0–39.0)	77	31.34（24–41）	25.32（15.1–36.9）	45	30.33（19–40）	24.91（18.0–38.0）
Physiological shunt abnormalities	1,005	30.7 (16–48)	29.2 (21.7–37.7)	63	33.38（26–49）	29.21（22.6–38.4）	31	30.29（23–38）	25.42（20.0–32.0）
Other structural abnormalities	687	30.8 (17–43)	28.1 (18.1–38.4)	53	32.45（21–43）	27.05（15.1–38.4）	56	30.11（21–38）	24.31（18.5–32.0）
Cardiac position abnormalities	273	31.0 (19–44)	27.2 (18.9–36.3)	-	-	-	2	25.50（23–28）	20.35（19.6–21.1）
Total (unique cases)	50,093	31.4 (16–54)	26.5 (16.0–41.1)	1458	32.2 (18–49)	25.93 (15.1–39.4)	3370	28.9 (19–49)	23.76（15.0–38.0）

Age and gestational age are reported as mean (range). A positive case may involve more than one disease family; family-specific counts therefore cannot be summed to obtain the number of independent cases. Totals exclude overlap between families. Dashes indicate that no cases in the corresponding family were observed during validation. The seven disease families comprise 73 disease entities. Neither validation cohort has patient-level overlap with FetalCardioDB or with the other validation cohort.

Table 2 | Distribution of disease families across five provinces in the multicenter validation cohort

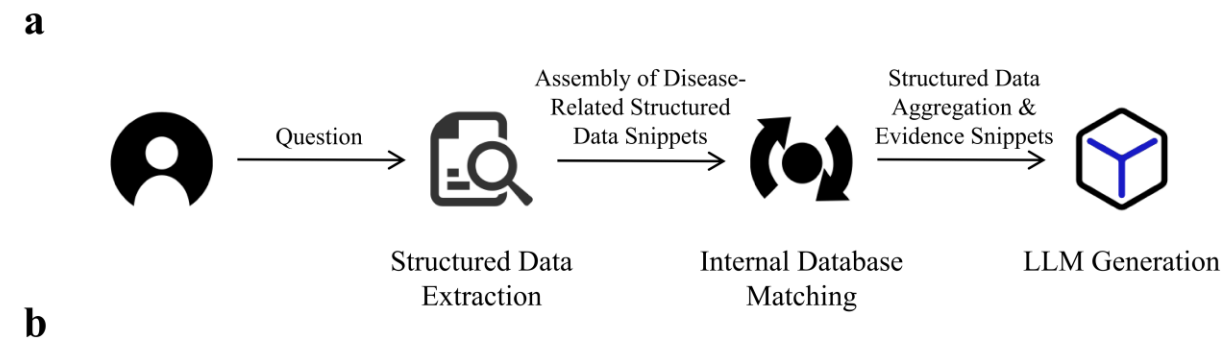
Diagnostic family	Total	Fujian	Ningxia	Guangdong	Anhui	Liaoning
Negative/normal	3,054	1,632	835	448	49	90
Septal defects	139	18	38	64	18	1
Great vessels/aortic arch	123	25	8	45	44	1
Venous return/systemic veins	34	8	3	11	11	1
Valves/outflow tracts	45	2	-	43	-	-
Physiological shunt abnormalities	31	4	9	14	4	-
Other structural abnormalities	56	7	7	40	2	-
Cardiac position abnormalities	2	-	-	2	-	-
Total (unique cases)	3,370	1,691	899	592	96	92

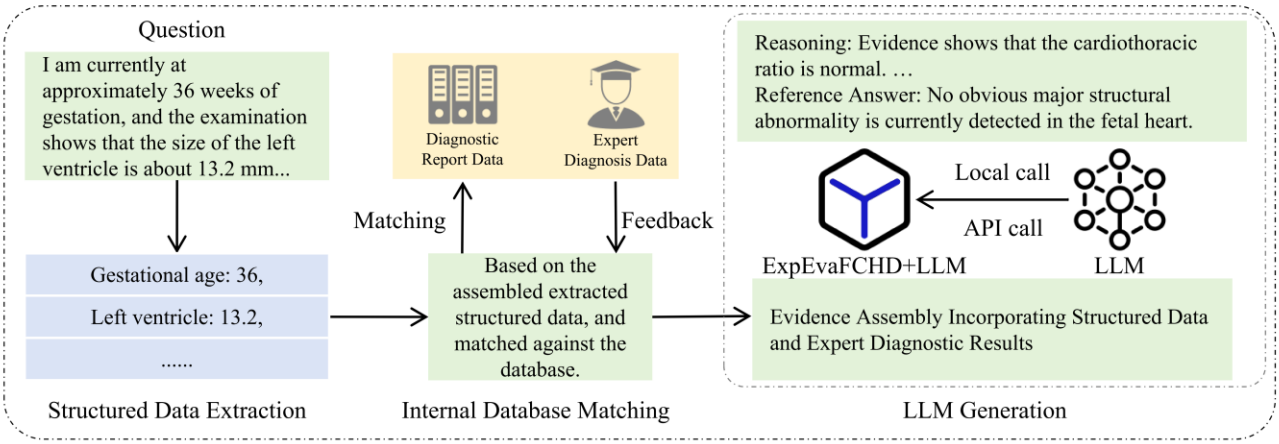
A case may involve more than one disease family. The final row reports independent case counts. Dashes indicate that no cases in the corresponding family were observed. The numbers of positive cases were 59 in Fujian, 64 in Ningxia, 144 in Guangdong, 47 in Anhui and 2 in Liaoning.

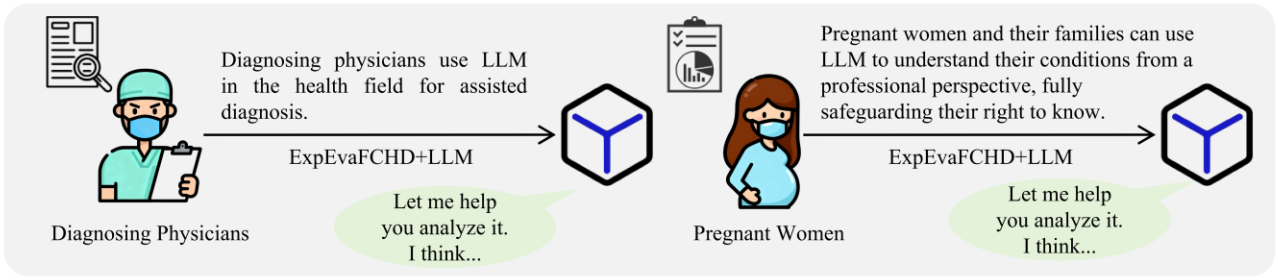
Framework overview and intended use in low-resourced settings

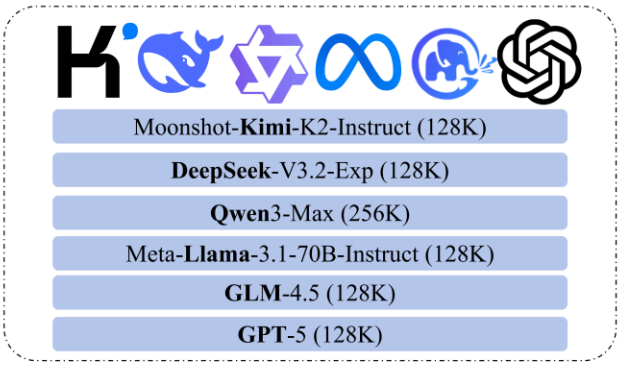
ExpEvaFCHD organizes screening into four stages: structured alignment, similar-case retrieval, polarity-controlled evidence aggregation and evidence-aligned constrained generation. The system directly analyzes structured fetal echocardiographic parameters measured by ultrasound equipment or already recorded in examination worksheets. This reduces the data-processing burden and reliance on large multimodal training datasets, while standardizing field names, measurement units and data formats improves adaptability to differences in documentation across institutions. By using existing clinical examination data, the design facilitates application in resource-limited settings. Figure 1 summarizes the overall architecture, clinical use pathways, six evaluated base-model backends and three-stage study design.

Within the clinical workflow, ExpEvaFCHD primarily provides frontline ultrasound physicians with diagnostic support based on structured measurements and local expert cases. The same diagnostic evidence can also be expressed in accessible language to help pregnant women and their families understand ultrasound findings. By linking abnormality-related statements to specific measurements and retrieved cases, the system makes their supporting evidence traceable and verifiable. Generated diagnostic reports undergo evidence validation and physician review to support clinical decisions and physician–patient communication.


a
Question
Assembly of Disease-Related Structured Data Snippets
Structured Data Aggregation & Evidence Snippets
Structured Data Extraction
Internal Database Matching
LLM Generation


b
Question
I am currently at approximately 36 weeks of gestation, and the examination shows that the size of the left ventricle is about 13.2 mm...
Gestational age: 36,
Left ventricle: 13.2,
.....
Structured Data Extraction
Diagnostic Report Data
Expert Diagnosis Data
Matching
Feedback
Based on the assembled extracted structured data, and matched against the database.
Internal Database Matching
Reasoning: Evidence shows that the cardiothoracic ratio is normal. …
Reference Answer: No obvious major structural abnormality is currently detected in the fetal heart.
Local call
API call
ExpEvaFCHD+LLM
LLM
Evidence Assembly Incorporating Structured Data and Expert Diagnostic Results
LLM Generation


c
Diagnosing physicians use LLM in the health field for assisted diagnosis.
ExpEvaFCHD+LLM
Let me help you analyze it. I think...
Diagnosing Physicians
Pregnant women and their families can use LLM to understand their conditions from a professional perspective, fully safeguarding their right to know.
ExpEvaFCHD+LLM
Let me help you analyze it. I think...
Pregnant Women


d
Moonshot-Kimi-K2-Instruct (128K)
DeepSeek-V3.2-Exp (128K)
Qwen3-Max (256K)
Meta-Llama-3.1-70B-Instruct (128K)
GLM-4.5 (128K)
GPT-5 (128K)
Base LLMs

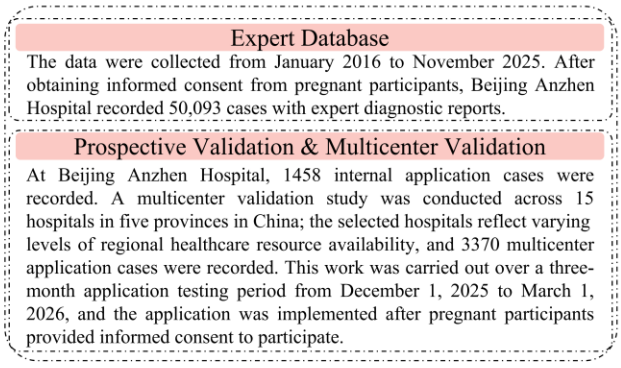

e
Expert Database
The data were collected from January 2016 to November 2025. After obtaining informed consent from pregnant participants, Beijing Anzhen Hospital recorded 50,093 cases with expert diagnostic reports.
Prospective Validation & Multicenter Validation
At Beijing Anzhen Hospital, 1458 internal application cases were recorded. A multicenter validation study was conducted across 15 hospitals in five provinces in China; the selected hospitals reflect varying levels of regional healthcare resource availability, and 3370 multicenter application cases were recorded. This work was carried out over a three-month application testing period from December 1, 2025 to March 1, 2026, and the application was implemented after pregnant participants provided informed consent to participate.
Datasets

Fig. 1 | System architecture, clinical use pathways and study design of ExpEvaFCHD. a, Structured measurement inputs, field alignment, similar-case retrieval, polarity-controlled evidence aggregation and constrained outputs. b, Institution-controlled local evidence processing and model-backend access. c, Physician-assisted interpretation and explanatory pathways for pregnant women and their families. d, The six model backends and their access configurations. Four open-weight models were accessed through the computing platform at the Hangzhou International Campus of Beihang University; Qwen3-Max and GPT-5 were accessed through test APIs provided by the research team. e, Case numbers, study periods and patient-level separation of FetalCardioDB and the two validation cohorts.

Prospective single-center validation shows improved report agreement

In prospective single-center validation (n = 1,458), ExpEvaFCHD significantly improved agreement between reports generated by all six base models and expert diagnostic reports. Among negative cases, cosine similarity (Cosine) increased from 0.824–0.844 to 0.939–0.958, CosineTI from 0.545–0.567 to 0.874–0.893, and Relevance from 0.817–0.837 to 0.965–0.978. Among positive cases, these three metrics increased from 0.693–0.720, 0.323–0.344 and 0.645–0.664 to 0.859–0.884, 0.735–0.763 and 0.826–0.844, respectively. Validity also increased from 0.325–0.348 to 0.496–0.517 among negative cases and from 0.260–0.284 to 0.469–0.494 among positive cases, indicating improvements in both semantic agreement and content validity (Supplementary Table S1 and Supplementary Fig. S1).

Informativeness decreased from 0.704–0.726 to 0.352–0.372 among negative cases and from 0.777–0.794 to 0.424–0.444 among positive cases. This change was consistent with the intended effect of evidence constraints: more focused generated content and fewer unsupported elaborations. Case-level paired analyses showed that changes in all five report-quality metrics in both validation cohorts remained statistically significant after Holm correction for multiple comparisons (all $P < 0.001$; Supplementary Table S2). Considered alongside the subsequent screening results, evidence-support audits and AI-assisted physician interpretation, these findings further support the role of ExpEvaFCHD in improving the consistency and standardization of diagnostic reports.

Case-level screening improves both exclusion and detection of abnormalities

ExpEvaFCHD significantly improved the detection and identification of abnormalities by all six base models in case-level FCHD screening. In the prospective cohort, sensitivity increased from 53.24%–73.24% with direct generation to 88.82%–92.94%, specificity from 24.69%–79.96% to 98.03%–98.93%, and overall accuracy from 36.01%–77.64% to 95.88%–97.46%. After enhancement, positive predictive value (PPV) and negative predictive value (NPV) reached 93.21%–96.32% and 96.65%–97.87%, respectively (Table 3, Fig. 2a,b and Supplementary Table S12).

These results indicate that ExpEvaFCHD can reduce false-positive findings in normal cases while improving the detection of fetal cardiac abnormalities. Integrating structured measurements with expert case evidence further strengthened the consistency and reliability of screening decisions. Case-level paired analyses showed improvements in sensitivity, specificity and accuracy for all six models relative to their base conditions (Supplementary Table S5 and Supplementary Fig. S3). Despite these overall gains, each enhanced model still produced 12–22 false-positive and 24–38 false-negative results, indicating the continued need to integrate system outputs with ultrasound findings and physician judgment in clinical practice to reduce the risks of missed and incorrect diagnoses.

Table 3 | Case-level screening performance of six base models with direct generation and the full ExpEvaFCHD system

Prospective validation: Base versus the full ExpEvaFCHD system (Full)

Model	Condition	Sensitivity	Specificity	PPV	NPV	Accuracy
Kimi-K2	Base	60.00% (54.71–65.07%)	78.80% (76.31–81.10%)	46.26% (41.66–50.92%)	86.63% (84.40–88.58%)	74.42% (72.12–76.59%)
Kimi-K2	Full	91.76% (88.36–94.24%)	98.48% (97.58–99.05%)	94.83% (91.88–96.75%)	97.52% (96.44–98.28%)	96.91% (95.90–97.69%)
DeepSeek-V3.2	Base	73.24% (68.29–77.66%)	24.69% (22.25–27.30%)	22.82% (20.43–25.41%)	75.20% (70.54–79.35%)	36.01% (33.58–38.51%)
DeepSeek-V3.2	Full	91.47% (88.02–94.00%)	98.66% (97.80–99.19%)	95.40% (92.55–97.19%)	97.44% (96.35–98.21%)	96.98% (95.97–97.74%)
GLM-4.5	Base	65.00% (59.79–69.88%)	72.00% (69.30–74.56%)	41.39% (37.28–45.61%)	87.12% (84.81–89.13%)	70.37% (67.98–72.66%)
GLM-4.5	Full	90.59% (87.02–93.25%)	98.21% (97.25–98.84%)	93.90% (90.77–96.02%)	97.17% (96.03–97.99%)	96.43% (95.35–97.27%)
Llama-3.1-70B	Base	60.00% (54.71–65.07%)	71.82% (69.12–74.38%)	39.31% (35.20–43.57%)	85.52% (83.12–87.62%)	69.07% (66.65–71.39%)
Llama-3.1-70B	Full	88.82% (85.03–91.75%)	98.03% (97.04–98.70%)	93.21% (89.93–95.47%)	96.65% (95.43–97.55%)	95.88% (94.74–96.79%)
Qwen3-Max	Base	53.24% (47.93–58.47%)	75.31% (72.70–77.75%)	39.61% (35.23–44.16%)	84.12% (81.72–86.25%)	70.16% (67.77–72.46%)
Qwen3-Max	Full	92.35% (89.03–94.73%)	98.93% (98.13–99.38%)	96.32% (93.68–97.88%)	97.70% (96.66–98.43%)	97.39% (96.44–98.10%)
GPT-5	Base	70.00% (64.93–74.63%)	79.96% (77.52–82.21%)	51.52% (46.96–56.04%)	89.76% (87.72–91.49%)	77.64% (75.43–79.71%)
GPT-5	Full	92.94% (89.71–95.21%)	98.84% (98.02–99.32%)	96.05% (93.36–97.68%)	97.87% (96.86–98.57%)	97.46% (96.52–98.15%)

Data are percentages with 95% Wilson confidence intervals. The prospective cohort included 1,458 cases (340 positive and 1,118 negative); the multicenter cohort included 3,370 cases (316 positive and 3,054 negative). Confusion-matrix counts, numerators, denominators and likelihood ratios are provided in Supplementary Table S12.

Multicenter validation: Base versus the full ExpEvaFCHD system (Full)

Model	Condition	Sensitivity	Specificity	PPV	NPV	Accuracy
Kimi-K2	Base	56.96% (51.45–62.31%)	78.00% (76.49–79.43%)	21.13% (18.52–23.99%)	94.60% (93.65–95.42%)	76.02% (74.55–77.44%)
Kimi-K2	Full	88.92% (84.99–91.93%)	97.71% (97.11–98.18%)	80.06% (75.56–83.90%)	98.84% (98.39–99.17%)	96.88% (96.24–97.42%)
DeepSeek-V3.2	Base	70.25% (64.99–75.03%)	23.90% (22.42–25.45%)	8.72% (7.68–9.88%)	88.59% (86.24–90.59%)	28.25% (26.75–29.79%)
DeepSeek-V3.2	Full	88.61% (84.63–91.66%)	97.90% (97.33–98.36%)	81.40% (76.94–85.15%)	98.81% (98.36–99.14%)	97.03% (96.40–97.55%)
GLM-4.5	Base	62.03% (56.56–67.20%)	71.19% (69.55–72.76%)	18.22% (16.02–20.63%)	94.77% (93.78–95.61%)	70.33% (68.76–71.84%)
GLM-4.5	Full	87.66% (83.57–90.84%)	97.41% (96.79–97.92%)	77.81% (73.21–81.82%)	98.71% (98.24–99.05%)	96.50% (95.82–97.07%)
Llama-3.1-70B	Base	56.96% (51.45–62.31%)	70.99% (69.35–72.57%)	16.89% (14.76–19.25%)	94.10% (93.06–94.99%)	69.67% (68.10–71.20%)
Llama-3.1-70B	Full	85.76% (81.48–89.18%)	97.18% (96.54–97.71%)	75.91% (71.21–80.06%)	98.51% (98.01–98.88%)	96.11% (95.41–96.71%)
Qwen3-Max	Base	50.32% (44.83–55.79%)	74.49% (72.92–76.01%)	16.95% (14.69–19.49%)	93.54% (92.50–94.45%)	72.23% (70.69–73.71%)
Qwen3-Max	Full	89.56% (85.70–92.47%)	98.10% (97.55–98.53%)	82.99% (78.64–86.61%)	98.91% (98.47–99.22%)	97.30% (96.70–97.80%)
GPT-5	Base	67.09% (61.73–72.04%)	79.21% (77.73–80.61%)	25.03% (22.23–28.06%)	95.88% (95.03–96.59%)	78.07% (76.64–79.44%)
GPT-5	Full	89.87% (86.05–92.74%)	98.00% (97.44–98.44%)	82.32% (77.94–85.98%)	98.94% (98.51–99.25%)	97.24% (96.63–97.74%)

Data are percentages with 95% Wilson confidence intervals. The prospective cohort included 1,458 cases (340 positive and 1,118 negative); the multicenter cohort included 3,370 cases (316 positive and 3,054 negative). Confusion-matrix counts, numerators, denominators and likelihood ratios are provided in Supplementary Table S12.

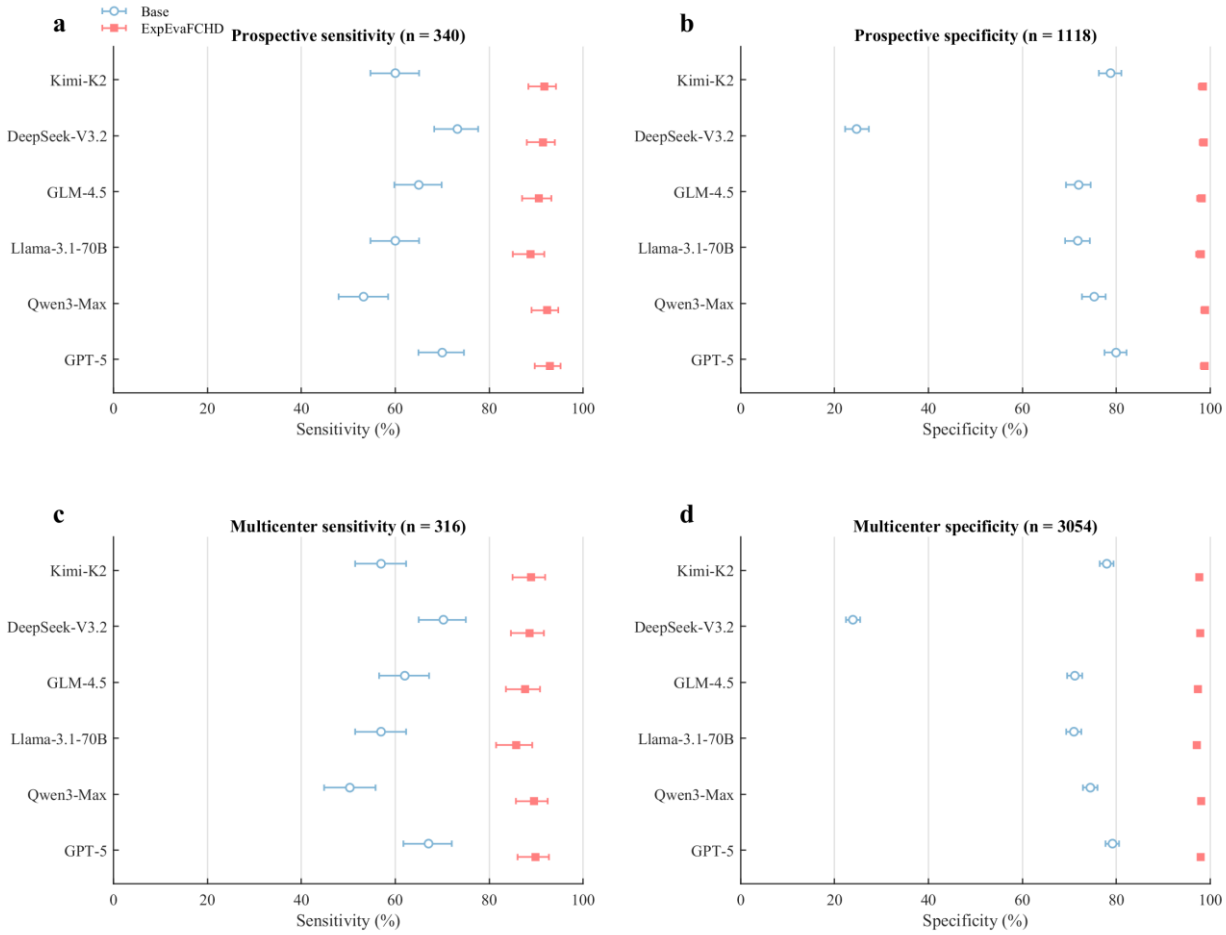


Fig. 2 | Screening sensitivity and specificity of six models in the prospective and multicenter cohorts. a,b, Sensitivity and specificity in the prospective cohort, based on 340 positive and 1,118 negative cases, respectively. c,d, Corresponding metrics in the multicenter cohort, based on 316 positive and 3,054 negative cases, respectively. Points indicate estimates; error bars represent 95% Wilson confidence intervals. Base denotes direct generation by the base model; Full denotes the complete ExpEvaFCHD workflow.

Comparator and ablation analyses support the contributions of structured evidence and validation

To assess the contributions of individual components to screening performance, we compared ExpEvaFCHD with gestational age-related rules, k-nearest neighbors (kNN), logistic regression, text RAG, structured RAG and several component-ablation conditions (Table 4, Fig. 3 and Supplementary Table S13). In the prospective cohort, the full Qwen3-Max system achieved sensitivity of 92.35% and specificity of 98.93%, exceeding text RAG (77.94% and 85.96%) and kNN (84.12% and 95.53%). This advantage was maintained in the multicenter cohort: sensitivity and specificity were 89.56% and 98.10% for the full system, 75.00% and 85.20% for text RAG, and 81.01% and 94.70% for kNN. These results indicate that structured-measurement retrieval and evidence validation can improve FCHD screening performance while maintaining stability across centers.

Ablation experiments further revealed a trade-off between abnormality detection and false-positive control. Removing the threshold constraint increased prospective sensitivity from 92.35% to 95.00%, but increased false-positive cases from 12 to 48. Removing evidence closure increased false-positive cases to 65. In the multicenter cohort, removing these two components increased false-positive cases from 58 with the full system to 156 and 202, respectively. Removing gestational age normalization or polarity control also affected sensitivity and specificity. These findings indicate that the components work together to maintain high abnormality detection while controlling false positives, improving reliability in practical screening settings.

To further assess the contribution of language generation, we compared the full Qwen3-Max system with template reports generated from identical retrieved evidence. Binary confusion-matrix counts were identical in both cohorts, with paired accuracy differences of 0 (Supplementary Table S6). However, readability and clarity scores were 0.063 and 0.064 higher for full-system reports in the prospective and multicenter cohorts, respectively, and clinical content completeness scores were 0.016 higher in both cohorts (Supplementary Table S16 and Supplementary Fig. S5). These findings further indicate that structured evidence processing provides key support for screening discrimination, while LLM generation helps organize diagnostic evidence into clearer, more complete and clinically usable reports.

Table 4 | Screening performance of the full Qwen3-Max system, comparators and component-ablation conditions

Prospective validation

Comparison condition	TP	FP	TN	FN	Sensitivity	Specificity	Accuracy
GA rules	272	67	1,051	68	80.00% (75.42–83.91%)	94.01% (92.46–95.25%)	90.74% (89.14–92.12%)
kNN	286	50	1,068	54	84.12% (79.85–87.62%)	95.53% (94.15–96.59%)	92.87% (91.43–94.08%)
Logistic regression	269	61	1,057	71	79.12% (74.48–83.10%)	94.54% (93.05–95.73%)	90.95% (89.36–92.31%)
Text RAG	265	157	961	75	77.94% (73.24–82.02%)	85.96% (83.80–87.87%)	84.09% (82.12–85.88%)
Structured RAG no validation	316	61	1,057	24	92.94% (89.71–95.21%)	94.54% (93.05–95.73%)	94.17% (92.85–95.26%)
No GA normalization	294	32	1,086	46	86.47% (82.42–89.70%)	97.14% (95.99–97.97%)	94.65% (93.37–95.69%)
No polarity control	310	58	1,060	30	91.18% (87.68–93.75%)	94.81% (93.35–95.97%)	93.96% (92.62–95.08%)
No SAWS threshold	323	48	1,070	17	95.00% (92.14–96.86%)	95.71% (94.35–96.75%)	95.54% (94.36–96.49%)
No closure	321	65	1,053	19	94.41% (91.44–96.39%)	94.19% (92.66–95.41%)	94.24% (92.92–95.32%)
kNN + template	286	50	1,068	54	84.12% (79.85–87.62%)	95.53% (94.15–96.59%)	92.87% (91.43–94.08%)
Matched evidence + template	314	12	1,106	26	92.35% (89.03–94.73%)	98.93% (98.13–99.38%)	97.39% (96.44–98.10%)
Full Qwen3-Max	314	12	1,106	26	92.35% (89.03–94.73%)	98.93% (98.13–99.38%)	97.39% (96.44–98.10%)
Direct Qwen3-Max	181	276	842	159	53.24% (47.93–58.47%)	75.31% (72.70–77.75%)	70.16% (67.77–72.46%)

TP, FP, TN and FN denote true positives, false positives, true negatives and false negatives, respectively. Parentheses following proportions indicate 95% Wilson confidence intervals. GA denotes gestational age; "No SAWS threshold" indicates removal of the score threshold constraint. Complete predictive values, likelihood ratios, numerators and denominators are provided in Supplementary Table S13; paired tests are provided in Supplementary Table S6.

Multicenter validation

Comparison condition	TP	FP	TN	FN	Sensitivity	Specificity	Accuracy
GA rules	243	208	2,846	73	76.90% (71.95–81.21%)	93.19% (92.24–94.03%)	91.66% (90.68–92.55%)
kNN	256	162	2,892	60	81.01% (76.33–84.95%)	94.70% (93.84–95.44%)	93.41% (92.52–94.20%)
Logistic regression	240	192	2,862	76	75.95% (70.94–80.33%)	93.71% (92.80–94.52%)	92.05% (91.09–92.91%)
Text RAG	237	452	2,602	79	75.00% (69.94–79.45%)	85.20% (83.90–86.41%)	84.24% (82.97–85.43%)
Structured RAG no validation	284	192	2,862	32	89.87% (86.05–92.74%)	93.71% (92.80–94.52%)	93.35% (92.46–94.15%)
No GA normalization	264	113	2,941	52	83.54% (79.06–87.22%)	96.30% (95.57–96.91%)	95.10% (94.32–95.78%)
No polarity control	279	183	2,871	37	88.29% (84.28–91.38%)	94.01% (93.11–94.80%)	93.47% (92.59–94.26%)
No SAWS threshold	291	156	2,898	25	92.09% (88.58–94.58%)	94.89% (94.05–95.62%)	94.63% (93.82–95.34%)
No closure	289	202	2,852	27	91.46% (87.85–94.06%)	93.39% (92.45–94.21%)	93.20% (92.30–94.01%)
kNN + template	256	162	2,892	60	81.01% (76.33–84.95%)	94.70% (93.84–95.44%)	93.41% (92.52–94.20%)
Matched evidence + template	283	58	2,996	33	89.56% (85.70–92.47%)	98.10% (97.55–98.53%)	97.30% (96.70–97.80%)
Full Qwen3-Max	283	58	2,996	33	89.56% (85.70–92.47%)	98.10% (97.55–98.53%)	97.30% (96.70–97.80%)
Direct Qwen3-Max	159	779	2,275	157	50.32% (44.83–55.79%)	74.49% (72.92–76.01%)	72.23% (70.69–73.71%)

TP, FP, TN and FN denote true positives, false positives, true negatives and false negatives, respectively. Parentheses following proportions indicate 95% Wilson confidence intervals. GA denotes gestational age; "No SAWS threshold" indicates removal of the score threshold constraint. Complete predictive values, likelihood ratios, numerators and denominators are provided in Supplementary Table S13; paired tests are provided in Supplementary Table S6.

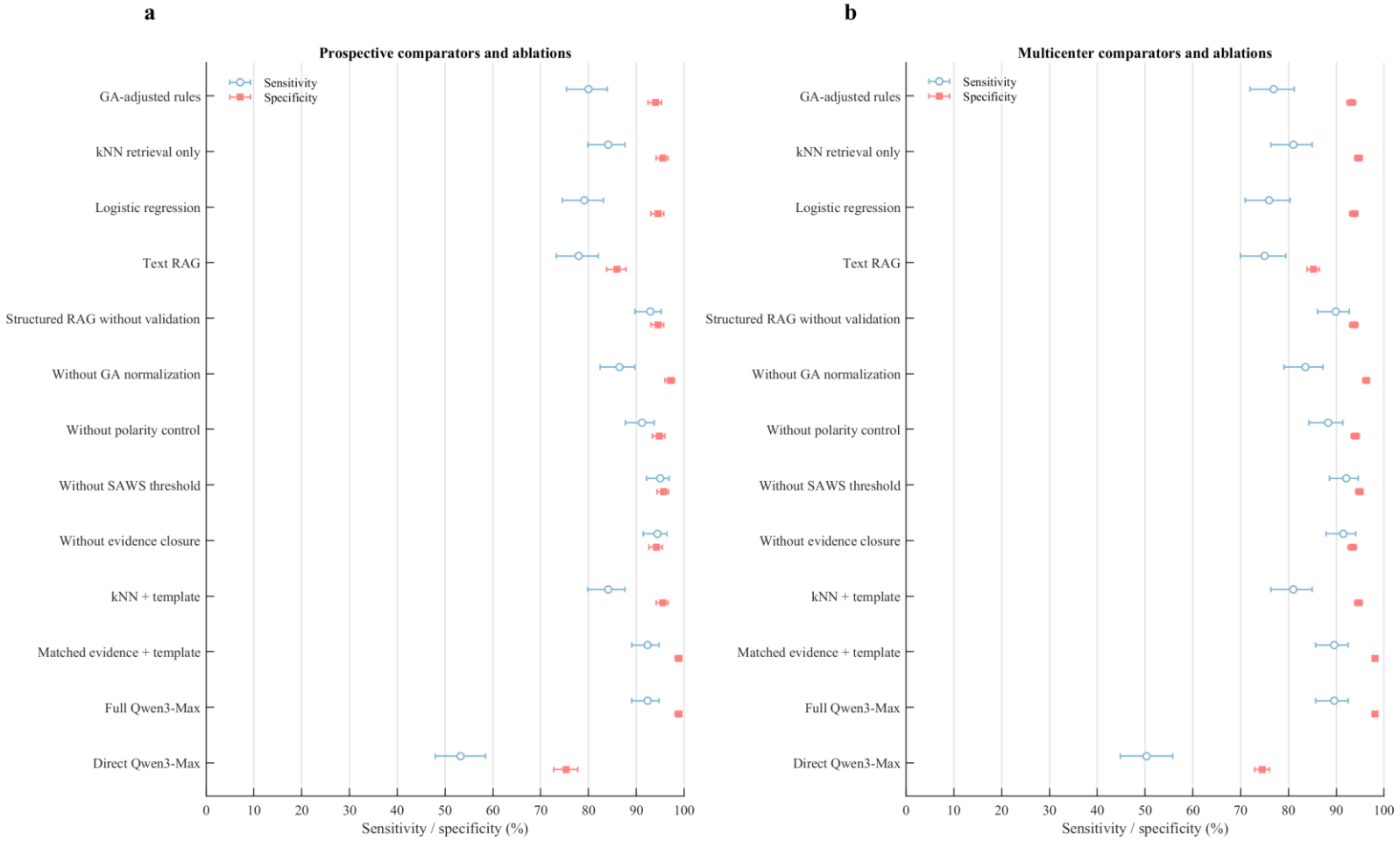


Fig. 3 | Screening performance of comparators and component-ablation conditions. a, Prospective cohort. b, Multicenter cohort. The comparisons include gestational age-related rules, kNN, logistic regression, text RAG, structured RAG without evidence validation, component-ablation conditions, template reports and the full Qwen3-Max system. Points indicate sensitivity or specificity estimates; error bars represent 95% Wilson confidence intervals. The matched-evidence template and full Qwen3-Max system yielded identical binary classification counts.

Multicenter validation preserves overall screening gains despite cross-center heterogeneity

In the multicenter validation cohort spanning 15 hospitals at different levels of care across five Chinese provinces (n = 3,370), ExpEvaFCHD maintained the screening advantages observed in prospective validation. After system enhancement, the six base models achieved sensitivity of 85.76%–89.87%, specificity of 97.18%–98.10% and overall accuracy of 96.11%–97.30% (Table 3 and Fig. 2c,d). For report quality, Cosine, CosineTI and Relevance reached 0.930–0.948, 0.863–0.882 and 0.954–0.971, respectively, among negative cases, and 0.851–0.872, 0.732–0.753 and 0.812–0.832 among positive cases. Consistent with prospective validation, all report-quality metrics improved to varying degrees except Informativeness, which decreased (Supplementary Tables S1 and S2 and Supplementary Fig. S2). These results indicate that the system maintained screening performance and diagnostic report agreement across centers.

Multicenter validation covered settings ranging from high-level specialist referral centers to regional hospitals and routine screening facilities. Eight hospitals primarily provided regional care, routine fetal cardiac screening and referral of complex cases to higher-level centers. Differences in specialist staffing, exposure to complex cases and clinical support allowed the evaluation to include real-world settings with relatively limited specialist resources. Validation was conducted within a common framework of structured measurements and expert reference diagnoses, providing a basis for further evaluation of ExpEvaFCHD under different healthcare resource conditions.

The multicenter cohort included hospitals with different resource levels and substantial variation in case composition and the proportion of positive cases. Positive-case proportions were 3.49%, 7.12%, 24.32%, 48.96% and 2.17% in Fujian, Ningxia, Guangdong, Anhui and Liaoning, respectively. In provincial analyses, sensitivity of the full systems ranged from 81.25% to 100%, and specificity from 92.22% to 100% (Fig. 4 and Supplementary Table S14). Sensitivity reached 100% in Liaoning, but the 95% confidence interval was wide (34.24%–100%) because only 2 positive cases were included. Despite regional differences in case mix, sample size and healthcare resources, ExpEvaFCHD maintained high overall screening performance, illustrating the adaptability of combining structured measurements with expert diagnostic evidence across institutions. Provincial variation also indicates that further implementation requires ongoing evaluation in relation to local case distributions and sample sizes.

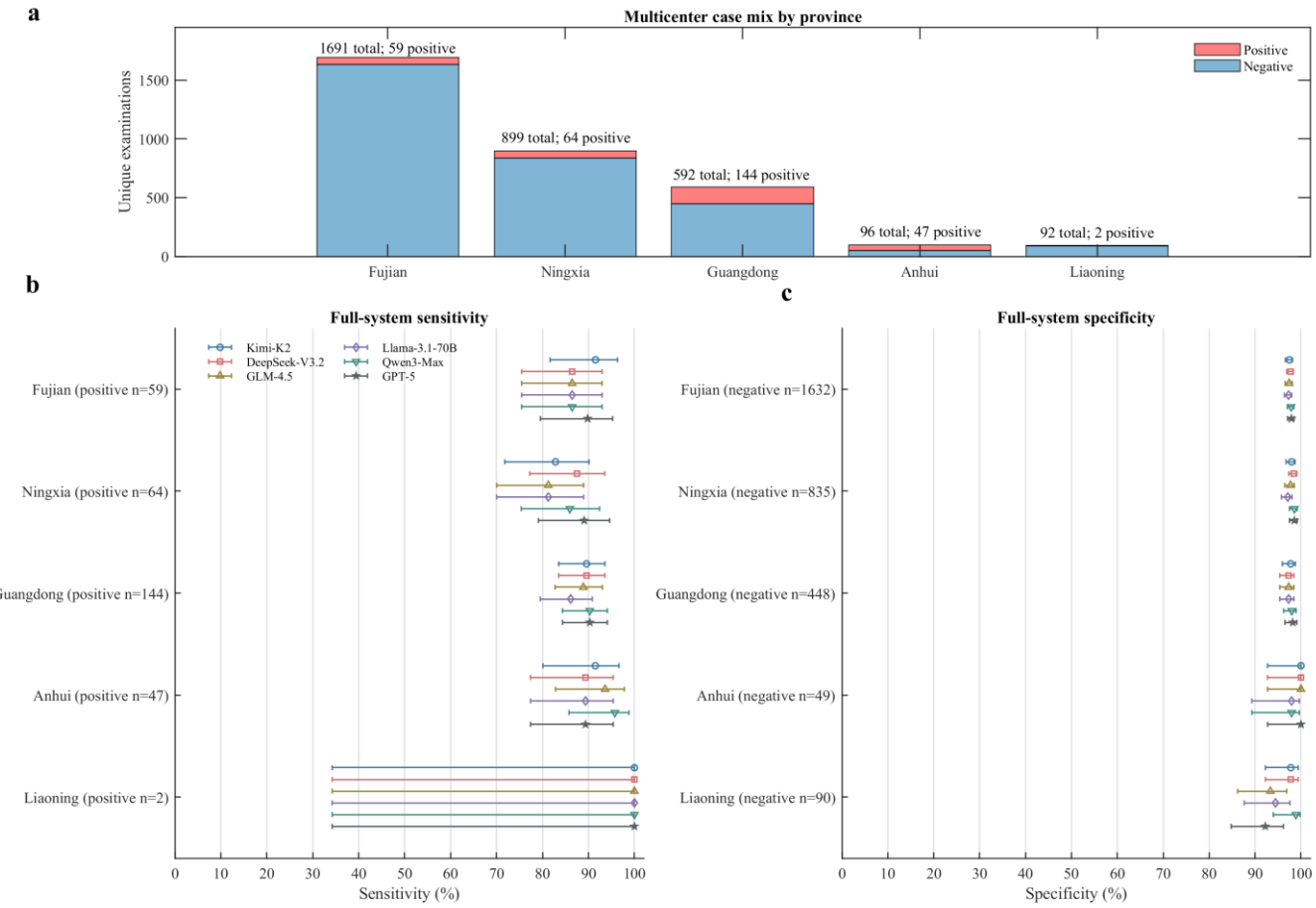


Fig. 4 | Provincial case composition and variation in screening performance in the multicenter cohort. a, Numbers of independent cases and their positive/negative composition in five provinces. b,c, Provincial sensitivity and specificity of the six full systems. Points indicate estimates; error bars represent 95% Wilson confidence intervals. Liaoning included only 2 positive cases, resulting in wide sensitivity intervals.

Disease-family set prediction improves differential diagnosis, including cases involving multiple families

When differential diagnosis was evaluated across seven disease families, ExpEvaFCHD significantly improved recognition of cardiac abnormalities in cases involving single and multiple families. Among positive cases in the prospective cohort, micro-averaged family recall increased from 9.29%–16.36% with the base models to 56.16%–63.03%, and micro-averaged precision increased from 46.22%–66.39% to 88.04%–91.95%. Mean Jaccard similarity increased from 0.075–0.190 to 0.537–0.632 for single-family cases and from 0.079–0.144 to 0.532–0.592 for multiple-family cases. Among positive cases in the multicenter cohort, micro-averaged family recall increased from 6.51%–13.72% to 54.42%–61.40%, and overall mean Jaccard similarity increased from 0.065–0.122 to 0.516–0.603 (Fig. 5 and Supplementary Table S4). These findings indicate that structured evidence aggregation helps distinguish different families of cardiac abnormalities and retain diagnostic information more completely in complex cases involving multiple abnormalities.

Across individual disease families, recognition was relatively strong for septal defects, with full-system recall reaching 71.22%–77.70% in the prospective cohort and 68.35%–74.82% in the multicenter cohort. Recall was lower for venous return and systemic venous abnormalities, at 36.17%–42.55% and 32.35%–41.18%, respectively, potentially reflecting their relative rarity and dependence on more detailed combinations of measurements. No reference-positive cases of cardiac position abnormalities occurred in the prospective cohort, precluding estimation of recall. The multicenter cohort included only 2 such cases, and all six enhanced models correctly identified 1 of them. Limited sample sizes for rare families continue to affect the stability of performance estimates and highlight the importance of expanding the representation of uncommon conditions in the expert case database. Family-specific counts, recall, specificity and precision are provided in Supplementary Table S3.

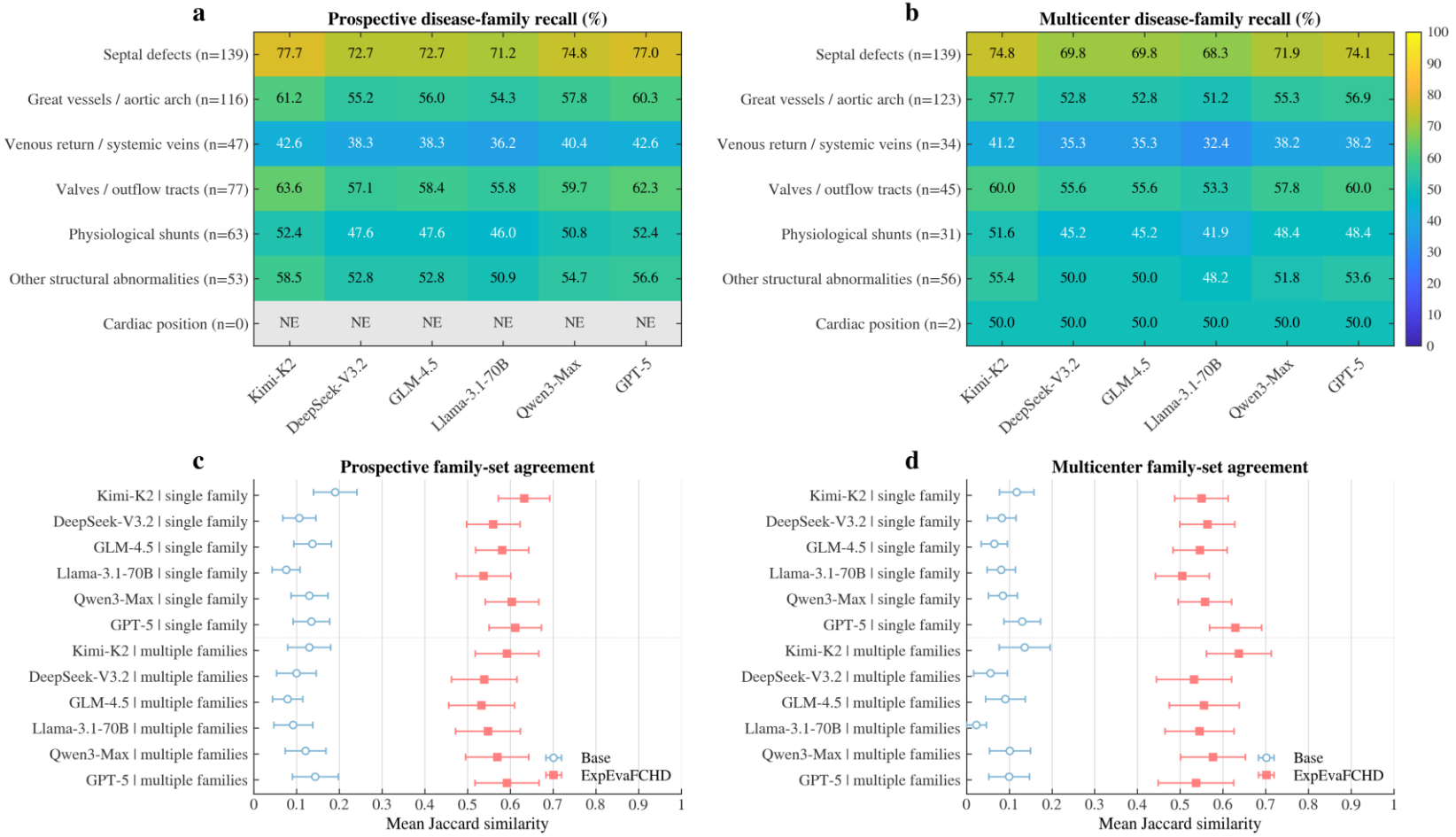


Fig. 5 | Recognition of seven disease families and agreement in family-set prediction. a,b, Recall for each of the seven disease families by the six full systems in the prospective and multicenter cohorts, respectively. Parentheses indicate the number of reference-positive cases for each family. No reference-positive cases of cardiac position abnormalities occurred in the prospective cohort, so recall could not be estimated. c,d, Mean Jaccard similarity and 95% confidence intervals for the base and full conditions in single-family and multiple-family cases in the two cohorts. Set prediction was evaluated at the level of the seven disease families.

Predictive values vary with prevalence, and calibration of the shared retrieval risk layer improves

The prospective and multicenter validation cohorts differed in the proportion of positive cases and showed corresponding differences in predictive values. PPV for the full systems was 75.91%–82.99% in the multicenter cohort, below the 93.21%–96.32% observed in the prospective cohort, whereas NPV reached 98.51%–98.94%. To evaluate the effect of disease prevalence on screening results, we examined changes in predictive values across assumed prevalences while holding each model's sensitivity and specificity constant (Fig. 6a,b,d,e). At an assumed prevalence of 1%, the full Qwen3-Max system had an estimated PPV of 46.50% (95% CI 33.10%–60.43%) and NPV of 99.922% (95% CI 99.887%–99.946%) based on the prospective results. Corresponding estimates based on the multicenter results were 32.26% (95% CI 26.91%–38.13%) and 99.893% (95% CI 99.852%–99.922%). These findings show that disease prevalence is an important determinant of screening predictive values and that local disease distributions should be considered when evaluating implementation in settings with different resources and case mixes. Complete predictive-value estimates across assumed prevalences are provided in Supplementary Table S18.

We further evaluated probability calibration in the shared retrieval risk layer. Compared with the raw evidence-support score, fixed probability mapping reduced the Brier score from 0.086 to 0.067 and expected calibration error (ECE) from 0.111 to 0.010 in the prospective cohort. Corresponding reductions in the multicenter cohort were from 0.091 to 0.043 and from 0.174 to 0.005. After mapping, calibration slopes were 0.968 (95% CI 0.859–1.077) and 0.903 (95% CI 0.819–0.987) in the prospective and multicenter cohorts, respectively, indicating closer agreement between estimated risk and observed outcomes. Grouped reliability results are shown in Fig. 6c,f and Supplementary Tables S7 and S8. These findings further support the use of structured evidence retrieval for risk assessment and provide a quantitative basis for application in populations with different disease prevalences.

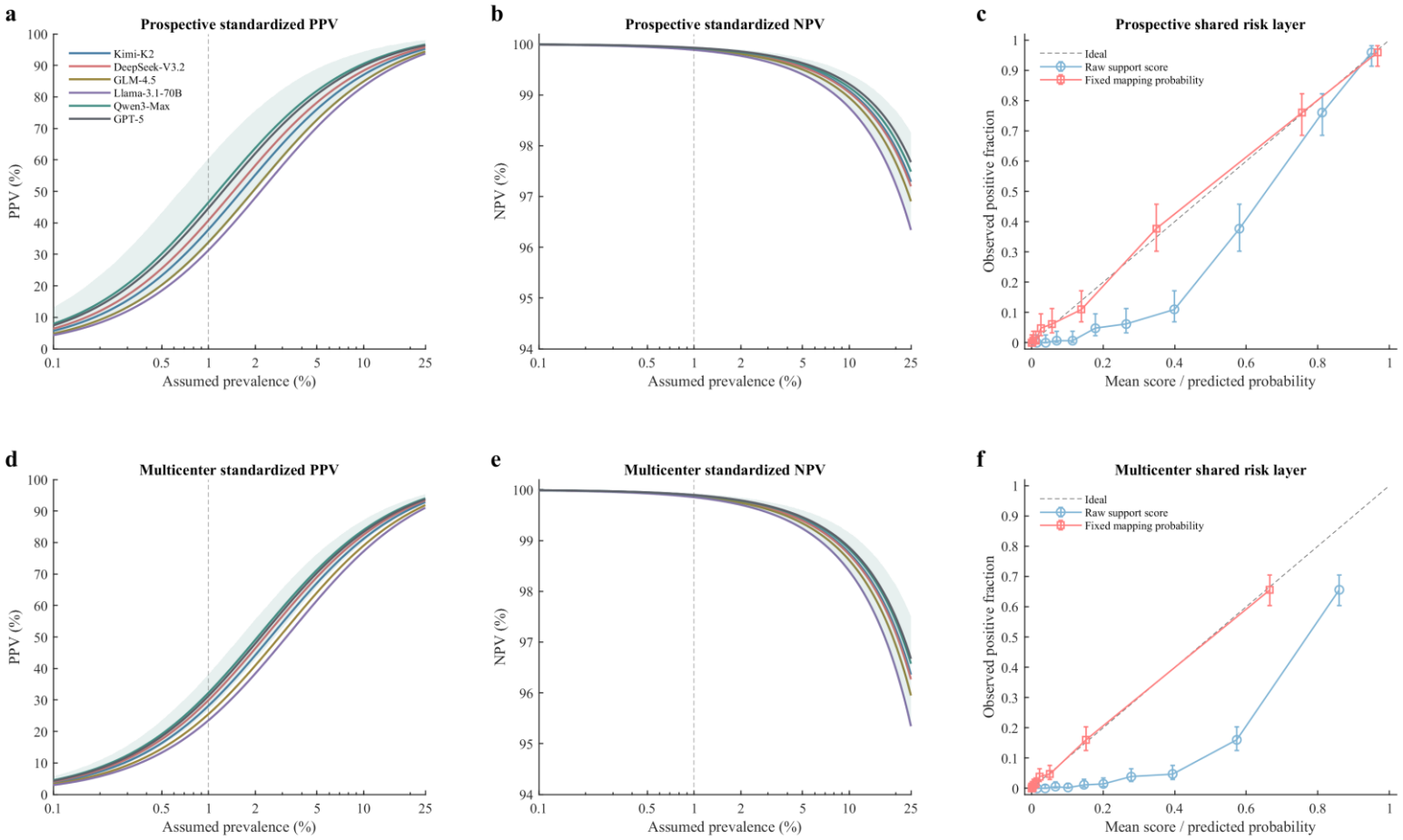


Fig. 6 | Predictive values across assumed prevalences and calibration of the shared retrieval risk layer. a,b, PPV and NPV estimated from prospective-cohort sensitivity and specificity. d,e, Corresponding curves based on the multicenter cohort. Assumed prevalence ranges from 0.1% to 25%; vertical dashed lines indicate 1%. Shading shows 95% confidence intervals for Qwen3-Max only, obtained by monotonic transformation of positive and negative likelihood-ratio intervals; uncertainty in target prevalence or transfer across populations is not included. c,f, Reliability plots for the shared retrieval risk layer, comparing raw support scores with fixed-mapping probabilities. Error bars represent 95% Wilson confidence intervals for the observed positive-case proportion in each group; diagonal dashed lines indicate ideal calibration.

AI-assisted physician interpretation improves screening and reduces report drafting and review time

In paired comparisons of unaided physician interpretation and ExpEvaFCHD-assisted interpretation, AI assistance significantly improved screening performance in both validation cohorts. In the prospective cohort, sensitivity increased from 85.59% to 93.82% and specificity from 96.87% to 98.57%; corresponding increases in the multicenter cohort were from 83.54% to 91.77% and from 96.30% to 98.00% (Table 5 and Fig. 7a,b). In the prospective cohort, 50 initially incorrect interpretations became correct after AI assistance, whereas only 3 changed in the opposite direction. Corresponding counts in the multicenter cohort were 97 and 19. Paired increases in case-level interpretation accuracy were 0.032 (95% CI 0.018–0.046) and 0.023 (95% CI 0.013–0.033), respectively; both were statistically significant (exact McNemar test, $P = 5.52 \times 10^{-12}$ and $P = 8.63 \times 10^{-14}$; Fig. 7d and Supplementary Table S15).

AI assistance also significantly reduced clinical report drafting and review time. Mean time decreased from 10.498 to 5.733 min in the prospective cohort and from 10.557 to 5.809 min in the multicenter cohort. Mean paired time savings were 4.766 min (95% CI 4.711–4.821) and 4.748 min (95% CI 4.711–4.784), respectively (Fig. 7c and Supplementary Table S15). These results indicate that ExpEvaFCHD can improve FCHD screening accuracy while reducing the time physicians need to complete diagnostic reports, providing further support for its integration into routine prenatal ultrasound workflows and its contribution to clinical efficiency.

Table 5 | Screening performance and reporting time with unaided and AI-assisted physician interpretation

Cohort	Condition	Sensitivity	Specificity	Accuracy	Report drafting and review min
Prospective	Unaided interpretation	85.59% (81.46–88.92%)	96.87% (95.68–97.74%)	94.24% (92.92–95.32%)	10.498 (10.346–10.650)
Prospective	AI-assisted	93.82% (90.74–95.93%)	98.57% (97.69–99.12%)	97.46% (96.52–98.15%)	5.733 (5.573–5.893)
Multicenter	Unaided interpretation	83.54% (79.06–87.22%)	96.30% (95.57–96.91%)	95.10% (94.32–95.78%)	10.557 (10.452–10.663)
Multicenter	AI-assisted	91.77% (88.22–94.32%)	98.00% (97.44–98.44%)	97.42% (96.83–97.90%)	5.809 (5.699–5.920)

Parentheses following proportions and times indicate 95% confidence intervals. Sample sizes were 1,458 in the prospective cohort and 3,370 in the multicenter cohort. Time savings are paired case-level differences and may differ slightly from the difference between displayed means because each mean was rounded separately. Complete confusion-matrix counts, predictive values, likelihood ratios and paired tests are provided in Supplementary Table S15.

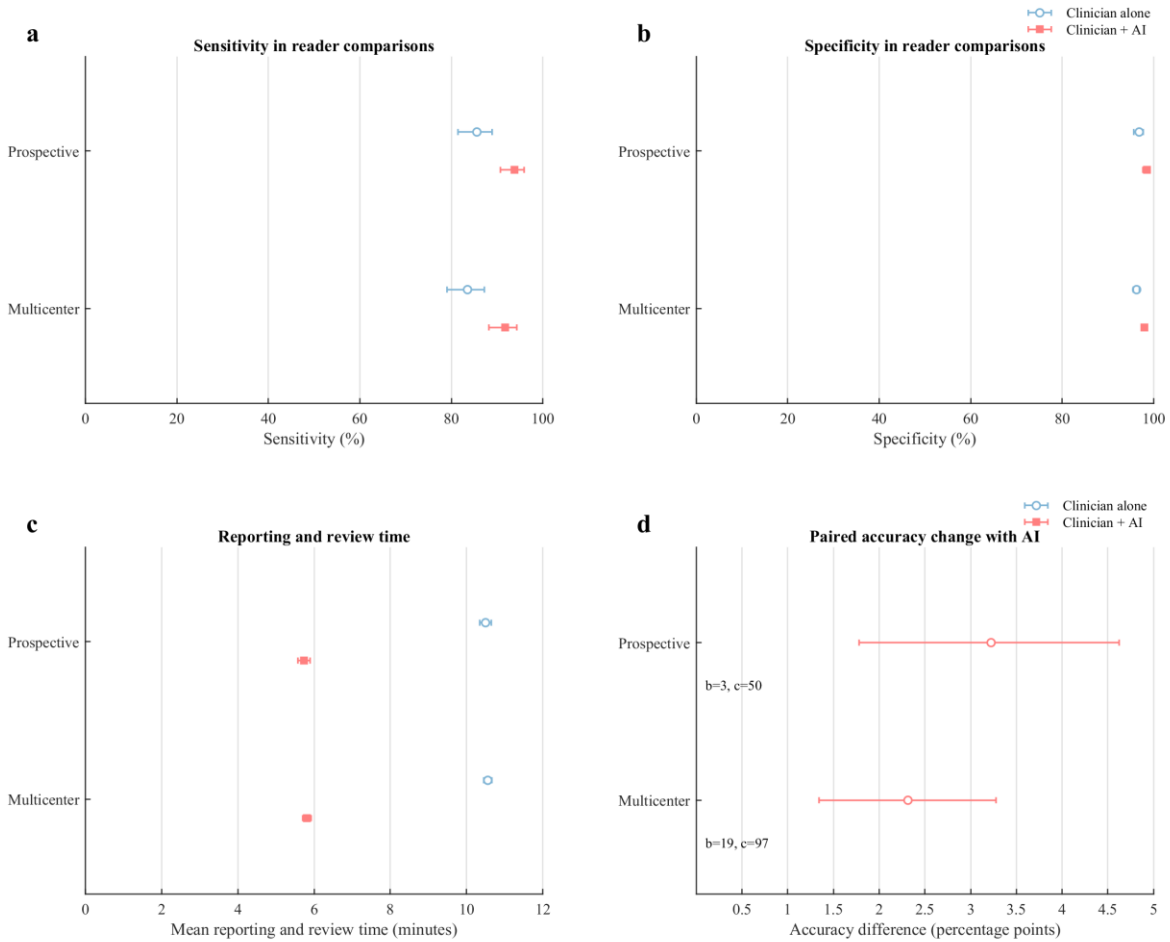


Fig. 7 | Screening performance and report completion time with unaided and AI-assisted physician interpretation. a,b, Sensitivity and specificity under unaided and AI-assisted interpretation in the two cohorts; error bars represent 95% Wilson confidence intervals. c, Mean report drafting and review time with 95% confidence intervals. d, Paired differences in case-level interpretation accuracy with conservative 95% confidence intervals constructed from simultaneous Clopper–Pearson bounds for the two discordant-pair proportions. Here, b denotes cases interpreted correctly without assistance but incorrectly with AI assistance, and c denotes the reverse. Counts were b = 3 and c = 50 in the prospective cohort, and b = 19 and c = 97 in the multicenter cohort. Interpretation errors included incorrect primary diagnoses and diagnostic omissions.

Statement-level evidence support and workflow outcomes during physician review

During daily final physician review, we audited the evidential support for abnormality statements in ExpEvaFCHD-generated reports. In the prospective cohort, the abnormality-statement support rate increased from 33.38%–34.78% with the base models to 95.42%–96.48% across the six enhanced models, and the unsupported abnormality-statement rate decreased to 3.52%–4.58%. Corresponding rates for the full systems in the multicenter cohort were 95.36%–96.44% and 3.56%–4.64% (Supplementary Table S11 and Supplementary Fig. S4). These metrics use abnormality statements as the unit of evaluation and reflect agreement between generated content, structured measurements and retrieved evidence. The results indicate that evidence constraints substantially improved report credibility and traceability, while physician review was retained to address the small proportion of unsupported abnormality statements.

Further analysis of workflow records for the full system using Qwen3-Max as the generation backend showed that 77.2% and 78.5% of reports were accepted unchanged or after minor revision in the prospective and multicenter cohorts, respectively. Major revision or rejection was required for 22.8% and 21.5%, while physician review or escalation events occurred in 30.8% and 23.9% of cases. Among false-negative cases, 24 in the prospective cohort and 23 in the multicenter cohort did not trigger escalation flags, representing 1.6% and 0.7% of all cases in the respective cohorts (Supplementary Table S9). Error-attribution analysis further showed extensive missingness in structured measurement fields in 9/26 prospective and 15/33 multicenter false-negative cases (Supplementary Table S10). These findings highlight the importance of input completeness for screening performance and indicate that auditable evidence records help locate potential errors and inform improvements to clinical review and referral-escalation mechanisms.

ExpEvaFCHD can also organize the same diagnostic evidence into different explanatory formats for different users. For physicians, the system emphasizes candidate lesions, relevant measurements, similar expert cases and referral suggestions. For pregnant women and their families, it can translate specialist diagnostic information into accessible, structured explanations to support understanding of ultrasound findings and the need for further examination or follow-up. By combining interpretation support with physician–patient communication, the design may help address specialist shortages and limited consultation time in low-resourced settings and support access to prenatal care. Report-quality scores and abnormality-statement evidence-support audits are presented in Supplementary Tables S16 and S11, respectively. The effects of patient-facing explanations on patient understanding and clinical communication outcomes require evaluation in subsequent studies.

Discussion

This study extends digital medicine for FCHD from model-centered technical development to clinical implementation: how can data already available in real-world ultrasound workflows be used to build clinically actionable and scalable diagnostic support? To address this question, we developed ExpEvaFCHD, which combines structured measurements obtained directly from ultrasound equipment with local expert diagnostic evidence through similar-case retrieval, evidence aggregation and constrained generation. The workflow supports prenatal screening in settings with limited specialist resources by using existing examination data and clinical processes, reducing reliance on large imaging datasets and base-model retraining. Our evaluation focused on clinical actionability, contextual adaptability and scalability, examining how expert diagnostic knowledge could be integrated into routine screening across healthcare resource levels and providing evidence for the transition from proof of concept to practical implementation.

The clinical actionability of ExpEvaFCHD is reflected in joint improvements in screening performance, report quality and physician efficiency. In the prospective cohort, the six enhanced models achieved sensitivity of 88.82%–92.94% and specificity of 98.03%–98.93%, alongside improvements in disease-family set recall and

diagnostic agreement in multiple-family cases. Paired physician comparisons connected these results to practical interpretation tasks: AI assistance increased physician sensitivity to 93.82% and 91.77% in the prospective and multicenter cohorts, respectively, and reduced report drafting and review time by approximately 4.8 min on average. Evidence-support audits further showed that most abnormality statements in enhanced-model reports could be traced to structured measurements or expert case evidence, allowing physicians to examine the basis for the reported findings.

Comparisons using identical evidence further clarified the roles of individual components. With Qwen3-Max, template reports and the full system yielded identical binary screening confusion-matrix counts, whereas full-system reports received higher scores for readability and clarity and clinical content completeness. This supports a division of function between structured evidence processing and language generation: expert case retrieval, evidence aggregation and validation underpin screening decisions, while the LLM organizes measurements and diagnostic evidence into clinically usable reports. The value of ExpEvaFCHD therefore lies in the complete process from evidence retrieval to report composition, bringing abnormality identification, evidence verification and report drafting into a single workflow.

Adaptation to the clinical context begins with the use of routine ultrasound data. Cardiac chamber dimensions, vessel diameters, Doppler velocities and cardiac axis are already documented in everyday examinations and can support expert case retrieval and analysis without retraining the base model for each generation backend. Based on our practical measurement experience, data acquisition should include standardized four-chamber and three-vessel views at a minimum to record basic chamber and great-vessel measurements. These views are relatively easy to obtain, have clearly defined measurement targets and readily documented parameters, and follow relatively uniform acquisition and measurement conventions, facilitating comparable structured data across levels of care. Clinical examinations should also include acquisition and assessment of other required views in accordance with routine fetal cardiac screening protocols. Building support around standard views and existing measurements can reduce additional data-curation work and facilitate integration into established workflows at resource-limited institutions.

Separating structured evidence processing from model-generation services also provides flexibility across infrastructure conditions. In the tested configuration, the four open-weight models—Kimi-K2, DeepSeek-V3.2, GLM-4.5 and Llama-3.1-70B—were accessed through the Data and Computing Platform at the Hangzhou International Campus of Beihang University. Qwen3-Max and GPT-5 were accessed through test API services provided by the research team, while participating hospitals independently performed physician interpretation and report review. The expert case database, field alignment, retrieval and evidence validation were organized separately from generation backends, allowing different models to share a common evidence-processing workflow. This modular design provides a basis for adapting model services, maintaining case databases and updating diagnostic rules according to institutional resources.

The multicenter study evaluated the workflow under different conditions of specialist-resource availability. The 15 participating hospitals across five Chinese provinces spanned four levels of healthcare service: high-level specialist referral centers, provincial diagnostic centers, regional hospitals and primary-level screening facilities. Some regional and routine screening hospitals had the basic capacity to perform prenatal ultrasound but relatively limited exposure to complex and rare fetal cardiac abnormalities, access to senior specialists and specialist consultation for difficult cases. The implementation question addressed here is how to supplement specialist diagnostic support within these existing examination settings. By bringing expert-consensus case evidence into frontline screening, ExpEvaFCHD provides institutions with different specialist capacities a means of drawing on expert knowledge.

Cross-center use also requires accommodation of differences in equipment, measurement fields, documentation practices and case composition. Through deterministic field alignment, ExpEvaFCHD retrieves expert cases

within a common measurement space, reducing the influence of field names, units and report wording on evidence matching. Despite substantial provincial differences in the proportion of positive cases, the six enhanced models maintained multicenter sensitivity of 85.76%–89.87% and specificity of 97.18%–98.10%. Provincial analyses also showed greater uncertainty where positive cases were sparse. Prevalence-standardized analyses demonstrated substantial changes in PPV across assumed prevalences, while fixed probability mapping improved calibration of the shared retrieval risk layer in the study data. These findings indicate that a structured-evidence workflow can support cross-institutional evaluation, with local case mix, prevalence and risk calibration remaining important considerations for wider implementation.

The user design of ExpEvaFCHD further connects assisted interpretation with clinical communication. For frontline ultrasound physicians in primary-level or non-tertiary institutions, the system can present candidate lesions, supporting measurements and similar expert cases to aid recognition of suspicious findings and consideration of further examination. The same evidence can be organized into accessible explanations for physicians to discuss findings and subsequent arrangements with pregnant women and their families. Clear evidence presentation and report wording have practical value where specialist consultation time is limited. The observed reductions in reporting time, improvements in readability and acceptance of most reports unchanged or after minor revision provide preliminary support for this use. Physician review remains integral to report use, addressing complex cases, revising inadequate statements and making final clinical judgments.

This work also builds on research in fetal medical AI and medical LLMs. AI research in fetal medicine has advanced ultrasound image recognition, standard-view acquisition, automated measurement and detection of congenital cardiac abnormalities.4[–13] Medical LLM research has progressed from general medical question answering to clinical reasoning, report generation and decision support, although work centered on structured prenatal ultrasound data and real-world screening workflows remains relatively limited.[22–26,29,32,33] ExpEvaFCHD uses expert-consensus cases as external evidence and combines them with current examination measurements through retrieval, validation and constrained generation to produce diagnostic interpretations. The contribution of this study is to organize these components into an auditable clinical support workflow and examine its performance in practical prenatal screening through prospective validation, multicenter evaluation across healthcare service levels and AI-assisted physician interpretation.

This study has several limitations. Input completeness and accuracy are important determinants of system performance. Error-attribution analysis showed that some false-negative cases had extensive missingness in key measurement fields. During testing, we also observed that the detail and diagnostic relevance of textual descriptions could influence generated outputs. Explicit measurements, abnormal findings and relevant clinical context provide richer case information, but textual descriptions were entered freely in this study. We did not uniformly score their completeness, accuracy or diagnostic value, nor systematically analyze the relationship between case-level input quality and model performance. The independent effect of input text quality on screening outcomes therefore remains unquantified, and these observations require further examination using prespecified information-quality assessments.

The current framework does not directly integrate static ultrasound images or video sequences, and structured retrieval cannot recover imaging findings that were never measured or documented. Recognition of some cardiac abnormalities depends on view-specific morphology, blood-flow patterns and spatial relationships between anatomical structures, which may be incompletely represented by existing measurements. Although the classification system covers seven disease families and 73 specific disease entities, set-based evaluation was conducted at the family level, and independent detection of individual entities remains insufficiently validated. Rare families were particularly underrepresented: no cardiac position abnormalities occurred in the prospective cohort, and only 2 occurred in the multicenter cohort, preventing stable performance estimates. Although the study covered different healthcare service levels, it was conducted entirely in China, and independent

performance across the four levels was not comprehensively compared. Applicability to other healthcare systems and populations requires further evaluation.

Residual workflow errors also warrant attention. Review still identified unsupported abnormality statements, and some false-negative cases did not trigger escalation flags, indicating scope to improve evidence validation, risk alerts and physician review. Evaluation focused on screening performance, report quality and report drafting and review time; referral appropriateness, patient understanding, counseling outcomes, follow-up adherence and patient outcomes were not directly measured. The use of a shared research computing platform and test APIs also means that hardware requirements, network dependence, data-transfer governance and long-term maintenance costs for independent operation at each hospital have not been systematically assessed. The fixed probability mapping in the shared retrieval risk layer also requires evaluation in independent data to establish transferability across populations. These factors will jointly influence sustained operation and clinical benefit in primary-level institutions.

To address variation in input information, future work could develop a guided data-entry framework that prompts users for information relevant to the current screening task, including gestational age, key chamber dimensions, vessel measurements and ultrasound findings. The framework could permit partial completion according to the examination performed and use missing-information detection and evidence-sufficiency assessment to prioritize additional parameters with high diagnostic value. Unavailable measurements would remain explicitly missing, with support based on the evidence available. This approach could reduce omissions and variation in free-text descriptions while retaining flexibility in resource-limited settings. Subsequent studies could compare the effects of different input formats on field completeness, generation quality and screening performance.

Multimodal extension is another important direction. Following de-identification and fulfillment of privacy requirements, routine ultrasound images, including four-chamber and three-vessel views, could be incorporated as evidence. One approach would use anatomical localization and automated measurement to convert image-derived morphological features and measurements into structured parameters or textual descriptions for the existing retrieval and evidence-aggregation workflow. Another would use vision models to analyze structural abnormalities and suspicious findings directly alongside existing measurement evidence. Both approaches could supplement information that is incompletely documented in the current inputs. Their incremental value for detecting complex and rare abnormalities should be evaluated across image quality, equipment conditions and healthcare service levels.

Continued expansion of expert-consensus cases, particularly rare and complex abnormalities, could also improve disease coverage within the retrieval space. Further work could refine field mapping, missing-data handling and review and escalation procedures according to local examination practices, alongside evaluation of deployment, data governance and costs under different resource conditions. Prospective studies could examine effects on referral arrangements, patient understanding and follow-up management. As structured evidence processing and integration of multiple evidence sources improve, the framework could also be explored for other assessments that rely on standardized measurements, including fetal growth restriction, maternal–fetal Doppler monitoring and assessment of cardiac valve and chamber function.

By integrating routine fetal ultrasound measurements with expert-consensus diagnostic evidence, this study established the ExpEvaFCHD workflow for settings with limited specialist resources. Prospective and multicenter results indicate improved screening performance and report agreement, together with reduced report drafting and review time during AI-assisted physician interpretation. Using existing examination data, separating evidence processing from generation backends and retaining physician review allows diagnostic support to be organized around actual clinical conditions. ExpEvaFCHD thus offers an actionable route for extending specialist knowledge to relatively resource-limited institutions and a basis for further evaluation of sustained digital medicine implementation and scaling.

Methods

Study design and ethics

The study comprised three stages: construction of a retrospective expert case database, prospective single-center validation and multicenter validation. ExpEvaFCHD uses a generation-enhancement strategy grounded in structured expert evidence, combining standardized and aligned fetal echocardiographic measurements with local expert case evidence to produce traceable diagnostic interpretations without updating base-model parameters. To assess overall performance and component contributions, we used direct generation by the same base models as the comparator and included rule-based screening, conventional classifiers, RAG, template reports and component-ablation experiments. A paired case-level design was also used to compare unaided and AI-assisted physician interpretation. ExpEvaFCHD primarily supports physicians in FCHD screening and diagnosis and can generate explanatory content for pregnant women and their families. Final diagnoses are made by physicians.

Reporting transparency was assessed against the TRIPOD+AI statement.[34] The assessment covers study populations, data sources, expert-consensus reference diagnoses, structured-input handling, model configuration, screening discrimination, calibration, clinical workflow outcomes and study limitations. A completed item-by-item TRIPOD+AI reporting checklist identifying manuscript and Supplementary Information locations, with explicit notes for partially reported or unavailable information, is provided as Supplementary Checklist 1.

The study was approved by the Medical Ethics Committee of Beijing Anzhen Hospital, Capital Medical University (approval no. KS2023025). All study procedures complied with the Declaration of Helsinki and applicable ethical standards. All participants provided written informed consent and received appropriate prenatal guidance and clinical care during the study. Research data were managed in accordance with privacy and data-security requirements, and directly identifying information was not used in the analyses. Participating hospitals conducted multicenter validation under ethics-approved bilateral collaboration arrangements and followed the blinding procedures specified in the study protocol. No deception was used, and there was no study-related direct interaction between the researchers or evaluating physicians and participants.

Across all study stages, gender was determined by participants' self-identification. We did not consider or preregister analyses stratified by sex or gender because these variables were considered theoretically unrelated to the research question.

Data sources, cohort separation, consensus labels and task taxonomy

FetalCardioDB comprised 50,093 fetal echocardiography cases retrospectively collected at Beijing Anzhen Hospital from January 2016 to November 2025. Prospective single-center and multicenter validation were both conducted from December 1, 2025 to March 1, 2026, enrolling 1,458 and 3,370 cases, respectively. The multicenter cohort came from 15 hospitals in five Chinese provinces and covered four levels of care: internationally leading specialist centers, provincial centers, regional hospitals and primary-level facilities. To evaluate applicability under different specialist-resource conditions, we classified the 15 hospitals by fetal cardiology service capacity and their roles in regional care pathways into high-level specialist referral centers (2 hospitals), provincial diagnostic and consultation centers (5), regional hospitals (5), and facilities primarily responsible for routine screening and referral of difficult cases to higher-level centers (3). The latter two groups comprised 8 hospitals that mainly provided routine fetal cardiac screening and had relatively limited exposure to complex and rare abnormalities and access to specialist consultation. This classification reflected differences in specialist service capacity and clinical workflows and provided a basis for evaluating the system in settings with relatively limited specialist resources. Strict temporal separation and patient-level isolation were applied.

Unique patient identifiers were used to remove duplicate records and ensure that the two validation cohorts had no patient-level overlap with FetalCardioDB or with one another. Repeat-visit and referral records for multicenter cases were also checked to prevent cross-institutional visits from introducing duplicate cases into the retrieval database.

Expert-consensus diagnoses served as the reference standard for all cases. Diagnostic labels in FetalCardioDB were jointly established by FCHD clinical experts, while multicenter reference diagnoses were adjudicated by consensus among fetal cardiology specialists, ultrasound physicians and pediatricians. The expert-developed taxonomy classified FCHD into seven disease families encompassing 73 specific disease entities. Cases with no FCHD identified were defined as negative, and cases with any FCHD entity were defined as positive. For family-level differential-diagnosis evaluation, the lesions in each case were mapped to these seven families to construct the expert-annotated and model-predicted disease-family sets, enabling assessment of recognition in single-family and multiple-family cases.

Study participants and intended users

Inclusion criteria for all cohorts were: (1) singleton pregnancy at 15–42 weeks of gestation with an indication for routine prenatal ultrasound; (2) outpatient or inpatient attendance for prenatal examination, willingness to cooperate with the examination and provision of informed consent; (3) a normal fetal heart rate of 110–160 beats per minute with no obvious arrhythmia; (4) absence of severe maternal comorbidities that could affect examination safety, such as preeclampsia, severe anemia or severe gestational diabetes; and (5) availability of clearly documented prenatal and postnatal follow-up information to support subsequent assessment of diagnostic agreement. Exclusion criteria were: (1) multiple pregnancy or other major fetal structural anomalies, such as severe hydrocephalus or spina bifida; (2) inability to cooperate with ultrasound examination or to complete simultaneous acquisition of imaging and positional information; (3) markedly unfavorable fetal position or excessive fetal movement causing unstable image acquisition; (4) very poor acoustic windows due to maternal body habitus or other factors, resulting in images unsuitable for analysis; (5) high-risk pregnancy with pre-existing maternal heart disease or a family history of genetic disease that could affect sample representativeness; and (6) refusal to provide informed consent or voluntary withdrawal during the study.

The intended users were: (i) frontline ultrasound physicians requiring support for FCHD screening and preliminary interpretation; and (ii) pregnant women and their families receiving structured, evidence-based explanatory outputs.

ExpEvaFCHD inference workflow

For an input examination $\tilde{\mathbf{x}}$, ExpEvaFCHD performs four stages, with structured evidence processing under local institutional control: (1) deterministic column-wise alignment to form a canonical structured vector $\mathbf{x}$; (2) similarity retrieval of comparable cases from the local case database $\mathcal{B} = \{(\mathbf{x}_i, y_i, \mathbf{z}_i, t_i)\}_{i=1}^{N}$, where $\mathbf{x}_i$ denotes aligned measurements, y_i the binary screening label, $\mathbf{z}_i$ the expert lesion set and t_i the expert diagnostic narrative; (3) polarity-controlled evidence aggregation over the retrieved set; and (4) evidence-aligned constrained generation by a fixed base LLM $\mathcal{M}$. Retrieval returns a top-k reference set $R_k(\mathbf{x})$, which is compressed into an evidence package $E(\mathbf{x})$ containing weighted abnormality cues, representative findings from similar cases and directly citable excerpts. Outputs comprise a binary screening conclusion $\hat{y}$, a lesion-set differential diagnosis $\hat{\mathbf{z}}$, a structured narrative report $\hat{r}$ and an attribution object a linking each abnormality statement to its supporting measurement fields or retrieved cases. The core design principle is evidence closure: abnormality-related statements must be supported by structured measurements or local case evidence. Compliance is assessed through subsequent validation and statement-level audits during physician review.

Structured alignment and normalization of heterogeneous inputs

ExpEvaFCHD takes a set of structured fetal echocardiographic parameters measured directly by ultrasound equipment or entered from examination worksheets, rather than raw image streams. Deterministic alignment rules handle aliases, unit differences, changes in field order, missing values and formatting irregularities. Values are converted to canonical units, synonymous fields are mapped to a common schema, missingness is encoded consistently, and clearly irrelevant free-text fragments are removed before downstream use. The aligned vector x is the sole structured signal permitted to drive retrieval and numerical consistency checks; unaligned raw text is not allowed to determine diagnostic conclusions. This design reduces cross-center vulnerability to naming differences, inconsistent units and heterogeneous reporting styles.

Similarity retrieval, source balancing and polarity-controlled evidence aggregation

To support real-world use when only some fields are available, similarity retrieval operates only on numerical fields present in the query, denoted by $J(\mathbf{x})$. For each available field j, aligned values are standardized using local-cohort statistics as follows:

$$z_j = \frac{x_j - \mu_j}{\sigma_j} \tag{1}$$

The nonnegative weighted distance is then defined as

$$d_w(\mathbf{x}, \mathbf{q}) = \sqrt{\sum_{j \in J(\mathbf{x})} w_j \left(z_j(\mathbf{x}) - z_j(\mathbf{q}) \right)^2} \tag{2}$$

and converted to a radial-basis similarity:

$$s(\mathbf{x}, \mathbf{q}) = \exp(-\gamma \, d_w(\mathbf{x}, \mathbf{q})^2) \tag{3}$$

Unless otherwise specified, retrieval returns the top-10 cases with similarity above a minimum threshold of 0.22. γ is initialized using a median-distance heuristic and can be adjusted for a particular deployment. If no cases remain after threshold filtering, the system falls back to the highest-similarity case as a low-similarity retrieval fallback. A fallback match does not itself establish sufficient support for an abnormality and remains subject to subsequent evidence validation. To reduce dependence on a single source distribution, an optional source-balancing mechanism groups retrieved cases by hospital and selects them in round-robin order. Evidence aggregation then extracts disease concepts and aliases from retrieved narratives and explicitly handles negation to prevent phrases such as “no abnormality,” “excluded” or “not observed” from being counted as positive evidence. A similarity-weighted positive-evidence proportion $r_{\mathrm{pos}}(x)$ is calculated from the retrieved set as a compact summary of abnormality support. The final evidence package contains ranked abnormality cues, concise summaries of positive or suspicious similar cases, and representative evidence excerpts for constrained generation.

Evidence-aligned constrained generation, attribution and deployment

The generation stage evaluated six base-model backends: Kimi-K2, DeepSeek-V3.2, GLM-4.5, Llama-3.1-70B, Qwen3-Max and GPT-5. All base-model parameters remained unchanged. The standardized structured measurement vector and retrieved evidence package were provided jointly to each model. Generation constraints ensured that measurement values remained consistent with the original records and required abnormality-related statements to have explicit support from structured measurements or similar-case evidence. Statement-level evidence support was subsequently checked using deterministic concept matching, numerical consistency checks and negation recognition. Draft reports that failed attribution requirements were rejected or handled through a fallback pathway. When no usable numerical evidence was available, the system returned an

insufficient-evidence report. If structured-output parsing failed, the system fell back first to section-level parsing and, if necessary, to a single recommendation text block, while retaining evidence validation. The abnormality-statement support rate evaluated agreement between generated reports and retrieved evidence; case-level diagnostic correctness was assessed separately using screening metrics.

In the tested configuration, the locally deployed Kimi-K2, DeepSeek-V3.2, GLM-4.5 and Llama-3.1-70B models were accessed through the Data and Computing Platform, Hangzhou International Campus, Beihang University. Qwen3-Max and GPT-5 were accessed through test API services provided by the research team. Model-access configurations are listed in Supplementary Table S17. The local expert case database, field alignment, similar-case retrieval and evidence validation were organized independently of the generation backends, allowing the base models to share a common structured evidence-processing workflow and to be evaluated using the same evidence.

Comparators, physician comparisons and workflow evaluation

To evaluate component contributions, comparators included gestational age-related rules, kNN, logistic regression, text RAG and structured RAG without evidence validation. We additionally compared kNN with a template, matched evidence with a template, the full Qwen3-Max system and direct generation by the base model. Component-ablation experiments separately removed gestational age normalization, polarity control, the SAWS threshold or evidence closure; all other parameters and experimental settings were held constant. All comparator and ablation conditions were evaluated in the same cohorts against the same case-level reference labels, with the full Qwen3-Max system as the paired comparison reference. The comparison between template and LLM-generated reports under identical evidence was used to distinguish the contributions of structured evidence processing and language generation. Readability and clarity and clinical content completeness were scored separately.

Each participating center independently conducted physician interpretation evaluations. Local physicians were responsible for checking diagnostic judgments, model performance and the reliability of ultrasound report content. In both validation cohorts, case-level results were recorded for unaided and AI-assisted interpretation, confusion matrices were calculated for each condition, and paired transition tables were constructed by matching interpretations of the same case. An incorrect primary diagnosis or a diagnostic omission was counted as a case-level interpretation error when assessing changes in diagnostic correctness with AI assistance. Report drafting and review times were also recorded under both conditions, with mean time and paired case-level time savings calculated in minutes.

Clinical workflow and error-attribution analyses were conducted for the full ExpEvaFCHD system using Qwen3-Max as the generation backend. Workflow events included acceptance unchanged, acceptance after minor revision, major revision or rejection, low-evidence triggers, evidence-validation failures, fallback after generation or formatting failures, physician review or escalation, and false-negative cases without escalation flags. Report acceptance was classified into mutually exclusive categories; other workflow events could co-occur and were analyzed separately. During daily final physician review, abnormality statements in model-generated reports were audited for evidential support to assess their consistency with structured measurements and retrieved case evidence.

Outcome measures

Report quality was evaluated using five metrics: cosine similarity (Cosine), TF–IDF cosine similarity (CosineTI), Relevance, Informativeness and Validity, reported separately for positive and negative cases. Relevance, Informativeness and Validity were calculated offline using an evaluation workflow based on GPT-2 and Vipu Baike resources to assess agreement with expert reference content and textual quality. Case-level screening

performance was evaluated using sensitivity, specificity, PPV, NPV, accuracy, and positive and negative likelihood ratios. Disease-family recognition was evaluated across the seven FCHD families using family-specific recall, specificity and precision. Agreement in disease-family set prediction was assessed using micro-averaged recall, precision and F1 score, mean Jaccard similarity and exact-match rate, with stratification by single-family and multiple-family cases.

Evidence support in generated reports was assessed using the abnormality-statement support rate (SFCR) and unsupported abnormality-statement rate (UASR). Both used the total number of abnormality statements as the denominator and quantified statements with and without supporting evidence, respectively. The proportion of all reports containing abnormality statements was calculated separately. Workflow event rates used the total number of cases in each cohort as the denominator. Error attribution was summarized separately for false-positive and false-negative cases using mutually exclusive categories. The shared retrieval risk layer was evaluated using the Brier score, ECE, calibration-in-the-large, calibration intercept and calibration slope for both raw evidence-support scores and fixed-mapping probabilities. Reliability analyses grouped predictions by decile and compared mean predicted probability with the observed positive-case proportion in each group; 95% confidence intervals for observed proportions were calculated using the Wilson method.

Statistical analysis

Descriptive statistics are reported as means and ranges or counts and proportions. Model outputs were parsed into structured screening labels and lesion sets using the same concept-extraction and negation-handling rules as evidence aggregation, producing reproducible confusion matrices for case-level analyses. The Wilson method was used to calculate 95% confidence intervals for sensitivity, specificity, PPV, NPV and accuracy. Confidence intervals for positive and negative likelihood ratios were calculated using the log delta method. The F_1 score was derived from precision P and recall R as $F_1 = \frac{2PR}{P+R}$. Paired binary comparisons between direct generation and ExpEvaFCHD-enhanced inference used the exact McNemar test, while paired continuous or ordinal report-quality metrics were compared using the two-sided Wilcoxon signed-rank test. Holm-adjusted P values were reported for multiple-model screening comparisons, comparator and ablation analyses, and report-quality comparisons as specified in the corresponding tables; physician comparisons reported exact-test P values separately. Disease-family set metrics were reported with 95% confidence intervals for single-family and multiple-family strata. Because ExpEvaFCHD also produces a similarity-weighted positive-evidence proportion $r_{\mathrm{pos}}(x)$, calibration was assessed using calibration-in-the-large, calibration slope, the Brier score and ECE calculated over deciles of $r_{\mathrm{pos}}(x)$, separating triage-risk calibration from the free-text narrative layer.

For the five report-quality metrics, 95% confidence intervals for means were calculated using Student's t distribution with n−1 degrees of freedom and standard error SD/√n, where n was the number of cases in the corresponding negative or positive stratum. Paired differences in screening performance between the base and full conditions were calculated as the difference between the two discordant-pair proportions. A 97.5% Clopper–Pearson interval was calculated for each proportion. The lower bound for improvement minus the upper bound for deterioration, and the upper bound for improvement minus the lower bound for deterioration, formed the difference interval. Conservative 95% intervals for paired differences in physician interpretation accuracy were likewise constructed using simultaneous Clopper–Pearson bounds for the two discordant-pair proportions. Continuous scores, times and disease-family set outcomes were reported with their corresponding estimates and intervals.

For prevalence-standardized analyses, sensitivity Se and specificity Sp were held constant within each validation cohort, and PPV and NPV were calculated at an assumed prevalence π as follows:

$$PPV(\pi) = \frac{\pi\, Se}{\pi\, Se + (1 - \pi)(1 - Sp)}$$

$$NPV(\pi) = \frac{(1 - \pi)\, Sp}{(1 - \pi)\, Sp + \pi(1 - Se)}$$

The plotted prevalence range was 0.1%–25%, with additional tabulated scenarios at 0.5%, 1% and 2%. Corresponding 95% intervals were obtained by monotonic transformation of positive and negative likelihood-ratio intervals: PPV increases with the positive likelihood ratio, whereas NPV decreases with the negative likelihood ratio. These analyses did not incorporate uncertainty in the target prevalence or transfer across populations and did not constitute an additional clinical validation cohort.

Data availability

The data are available from the corresponding authors upon reasonable request, subject to applicable regulatory and ethical requirements and the necessary approvals for use.

Code availability

Code is available at https://github.com/rzLbuaa/ExpEvaFCHD.git.

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Author contributions

Y.Z., R.L. and Y.G. contributed equally to this work. Y.H. and H.Z. jointly supervised the study. Y.Z. and R.L. conceived and designed the study. Y.Z., R.L., Y.G., H.S., J.W. and J.G. curated the data and developed the methodology. X.H. and Y.G. conducted the experiments and statistical analyses. X.G. and J.H. coordinated prospective and multicenter validation. Y.Z., Y.G. and J.W. participated in clinical interpretation and diagnostic adjudication. Y.Z. and R.L. drafted the manuscript. All authors reviewed, revised and approved the final manuscript.

Competing interests

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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